Cisco CCNA 200-301 - The Complete Guide To Getting Certified

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Contents

Chapter 1

Subnetting

1.1 Introduction

- Last section was about OSI layer 2, IP addresses, subnetmasks, etc.
- In this section we are still in layer 3 and we are going to go much deeper, and learn to control the boundaries.

1.2 CIDR - Classless Inter-Domain Routing

- CIDR - Classless Inter-domain routing:
  - A problem with classful addresses was that if a company had more than 254 hosts they would need to be assigned a class B network.
    - The internet authorities gave class A,B,C,D,E addresses. If they had 500 hosts they would get addresses for 65,543 hosts because the next class would allocate the whole next octet of the host portion. This was a waste.
  - They would have much less than the 65,434 hosts allocated, so this wasted a huge amount of the global address space.
  - Classless inter-domain routing (CIDR) was introduced in 1993 to alleviate this problem.
    - This removed the problem of classful addresses only allocating octets and instead allowed them to be split into smaller networks.
  - CIDR removed the fixed /8, /16, and /24 requirements for the address classes, and removed them to be split or 'subnetted' into smaller networks. For example 172.10.10.0 /20.
  - Companies can now be allocated an address range which more closely matches their needs and does not waste addresses.
  - The name for this is **subnetting**.
  - The benefit of Subnetting is that we don't waste addresses.
- CIDR and Route summarisation:
  - Another benefit of CIDR is that aggregate blocks of networks can be advertised on the internet.
  - For example lets say that a company called 'ISP A' gets assigned from the internet authorities the range of IP addresses ranging from 175.10.0.0/24 - 175.10.255.0. We also have another company called 'ISP B' are given 175.11.0.0/24 - 175.11.255.0/24.
  - Now these companies connect to each other and when CIDR comes in that ISP A and ISP B would need to copy all the routes from its routers and advertise all their 256 address blocks to each other, but when we have CIDR we just summarize the bits that they have in common so that we dont have duplicate data in the routing tables.

- In this case ISP A would advertise 175.10.0.0/16 to ISP B, and now ISP B only learns one router to get to ISP A rather than learning 256 routes. This is complicated and vaguely explained in this lecture, so watch the youtube video below.
- This was actually not very well explained in the course so this youtube video actually cleared it up: `https://www.youtube.com/watch?v=YOpwKwol8xk`

- Route summarization benefits:
  - ISP A does not know about all 256 /24 networks reachable in ISP B.
  - It only has the single 175.11.0.0/16 summary route. It has 1 route instead of 256. Less memory is used.
  - This reduces the size of ISP A's routing table and takes up less memory.
  - If an individual link goes down in ISP B, it has no impact on ISP A. The single summary route does not change.
  - (Routers in ISP B would have to recalculate their routing table if a link went down)
  - This restricts issues to the local part of the network and reduces CPU overload.

1.3 Subnetting overview

- Subnetting:
  - Subnetting is one of the core knowledge areas in networking, crucial for the CCNA and for networking in general.
  - To understand this lecture, think about it from the point of view of the originally intended IP4 design again, where all hosts which can communicate on the internet have a public IP address.
    - Think of it 10-15 years ago when people did not have problems with IP address shortage.
  - Let's say we're the network designer fro a small business with four departments spread over two offices, and we want to manage our own public address space.
    - We could buy 4 separate class C networks for that. But the problem is that public IP addresses are expensive.
  - Rather than purchasing separate address ranges for the different departments, we can purchase a single class C range and subnet it into smaller portions. And then we can subnet that to the different departments of our network.

- Borrowing host bits:
  - Let's say we've been allocated class C 200.15.10.0/24.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

  - To subnet the network into smaller subnets, we need to 'borrow' host bits and add them to the network portion of the address.
    - We are going to move the border of the subnetmask to the right.
    - WHEN WE ARE DOING SUBNETTING THE LINE ALWAYS MOVES TO THE RIGHT.
  - The network address line always moves to the right when we subnet.
  - The further to the right we go, the more subnets we'ññ have of that size but less hosts.

- To calculate the number of networks:
  - To calculate the number of available subnets, the formula is $2^{\text{subnet-bits}}$.

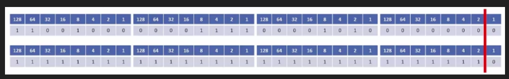
- If a class C network uses a 28 subnet mask then we've borrowed 4 bits from the default of /24.
- $2^4 = 16$ available subnets.
- If a class B network uses a /28 subnet mask then we've borrowed 12 bits from the default of /16.
- $2^{12} = 4096$ available subnets.
- Hosts on different subnets need to go via a router if they want to communicate with each other.

- To calculate the number of hosts:
  - To calculate the nuber of available hosts, the formula is $2^{\text{host-bits}} - 2$.
  - We subtract 2 because the network address and broadcast address cannot be assigned to hosts.
  - If a class C network uses a /28 subnet mask then we have 4 bits left for hosts.
  - $2^4 - 2 = 14$
  - If a class B network uses a /28 subnet mask then we have 4 bits left for hosts.
  - The amount of hosts we get is going to be dependant on the subnet size.

- A quick note on `ip subnet-zero` command:
  - Just like we have to subtract 2 to get the number of valid hosts, we used to have to subtract 2 to get the number of available networks also.
  - In the original internet standards, it was not allowed to use network bits of all 0's or all 1's (just like we can't use all host bits of all 0's or all 1's).
  - There wasn't really any practical need for this and it wasted address space.
  - The `ip subnet-zero` command on a router overrides the limitation, and is enabled by default.
  - This is important because on the updated standards we don't use the 2 subnets we use to take off for the network, we ONLY DO THIS FOR THE HOSTS. Hence, you DO NOT SUBTRACT 4, you subtract 2. Do not do that for the CCNA.

1.4 Subnetting classs C networks and VLSM

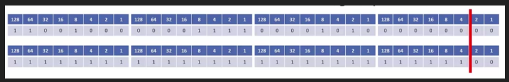
- Class C /31 subnet:
  - Let's say we've been allocated class C 200.15.10.0/24.



    * If we want to have any hosts available then the highest subnetmask we can have is /31. In that case we leave 1 bit remaining for the hosts.
  - If we move the line all the way to the right we're now using /31 (or 255.255.255.254).
  - This leaves one bit for the host address, with a possible value of 0 or 1.
  - It borrows 7 bits from the network address.
  - This gives us 128 subnets 2^7 which accommodates 2 hosts each.
  - We subnet using /31. Valid host addresse:
    * 200.15.10.0 to 200.15.10.1
    * 200.15.10.2 to 200.15.10.3
    * etc. to:
    * 200.15.10.254 to 200.15.10.255
  - Notice that the first three octets never change. The subnet mask changes in the first 7 bits.

  - What about the network and broadcast address?
    - /31 breaks the standard rules of IP addressing.
    - /31 subnets are supported in cisco routers for point to point links (which have no need for a network or broadcast address).
      - A point to point link means that all traffic moves from one to the other in terms of all messages sent can only be sent to the one other host in that subnet. There is no other place for the message to go so this is why we don't need a broadcast and network address.
      - This is why cisco allowed that exception, we can have a /31 subnet mask on a point to point link.

- Class C /31 subnet mask:
  - Let's say we've been allocated a class C 200.15.10.0/24.
  - Let's move the line back a place. We're now using /30 (or 255.255.255.252).
  - This leaves 2 bits for the host address, $2^2 - 2 = 2$ hosts available and the network and broadcast address.
  - It borrows 6 bits for the network address.
  - This gives us 64 subnets 2^6 which accommodate 2 hosts each.
  - Notice that the line is after the 4. The network addres goes up in values of 4.



  - Valid hosts addresses:
    - 200.15.10.1 to 200.15.10.2 (network .0, broadcast .3).
    - 200.15.10.5 to 200.15.10.6 (network .4, broadcast .7).
    - etc. to:
    - 200.15.10.253 to 200.15.10.254 (network .252, broadcast .255).
  -

1.4.1 /31 vs /30

- /31 and /30 both accommodate 2 hosts per subnet.
- /31 supports 128 subnets, /30 only 64.
- /31 us useful if you need to maximize use of your address space.
- /30 is more standard and commonly used.
- For the CCNA exam, use /30 when a subnet to support 2 hosts is required, unless told to use /31.
  - Because usually a /30 is the correct answer.

1.4.2 Class C /29 subnet

- Let's say we've been allocated Class C 200.15.10.0/24.
- Let's move the line back a place. We've now using /29 (or 255.255.255.248).
- This leaves 3 bits for the host addresses, $2^3 - 2 = 6$ possible hosts.
- It borrows 5 bits for the network address.
- This gives us 32 subnets 2^5 which accommodate 6 hosts each.